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Solution:

a) $y[n] = y[n - 2] + 2x[n] + 3x[n - 1]$

- **Linearity:**

Define a new designated output $z[n] = y[n] - y[n - 2]$. Equation (a) can be rewritten as

$$z[n] = 2x[n] + 3x[n - 1].$$

To test linearity, let's consider $x_1[n]$, $x_2[n]$ as two inputs and their corresponding outputs as $z_1[n] = 2x_1[n] + 3x_1[n - 1]$, $z_2[n] = 2x_2[n] + 3x_2[n - 1]$. For another input $x_3[n] = \alpha x_1[n] + \beta x_2[n]$, the output should be $z_3[n] = \alpha z_1[n] + \beta z_2[n]$ for a linear system. Inserting this into the system definition,

$$\begin{aligned} z_3[n] &= 2x_3[n] + 3x_3[n - 1] \\ &= 2(\alpha x_1[n] + \beta x_2[n]) + 3(\alpha x_1[n - 1] + \beta x_2[n - 1]) \\ &= \alpha z_1[n] + \beta z_2[n], \end{aligned}$$

confirming that the system is **linear**.

- **Time invariance:**

For an input-output pair $x[n]$ and $z[n]$, the system is time-invariant if for a delayed input $x'[n] = x[n - n_0]$, the output $z'[n]$ is equal to $z[n - n_0]$.

$$\begin{aligned} z'[n] &= 2x[n - n_0] + 3x[n - n_0 - 1] \\ &= z[n - n_0], \end{aligned}$$

confirming **time invariance**.

- **Causality:**

Since the output $y[n]$ doesn't require future values of input $x[n]$, the system is **causal**.

b) $y[n - 2] + 5y[n] = \sin\left(\frac{\pi}{3}n\right)x[n]$

- **Linearity:**

Define a new designated output $z[n] = y[n - 2] + 5y[n]$. The equation can be rewritten as

$$z[n] = \sin\left(\frac{\pi}{3}n\right)x[n].$$

Similarly, consider $x_1[n]$, $x_2[n]$ as two inputs and their corresponding outputs as $z_1[n]$, $z_2[n]$. For another input $x_3[n] = \alpha x_1[n] + \beta x_2[n]$,

$$\begin{aligned} z_3[n] &= \sin\left(\frac{\pi}{3}n\right)(\alpha x_1[n] + \beta x_2[n]) \\ &= \alpha z_1[n] + \beta z_2[n], \end{aligned}$$

confirming that the system is **linear**.

- **Time invariance:**

Consider $z'[n]$ as output corresponding to $x'[n] = x[n - n_0]$, we have

$$z'[n] = \sin\left(\frac{\pi}{3}n\right) x'[n] = \sin\left(\frac{\pi}{3}n\right) x[n - n_0], \text{ and}$$

$$z[n - n_0] = \sin\left(\frac{\pi}{3}(n - n_0)\right) x[n - n_0] \neq z'[n]$$

showing that the system is **time-varying**.

- **Causality**

Since the function $y[n]$ doesn't require future values of $x[n]$, the system is **causal**.

c) $y[n] = x[|n| - 1]$

- **Linearity:**

Consider $x_1[n]$, $x_2[n]$ as two inputs and their corresponding outputs as $y_1[n]$, $y_2[n]$. For another input $x_3[n] = \alpha x_1[n] + \beta x_2[n]$,

$$\begin{aligned} y_3[n] &= x_3[|n| - 1] \\ &= \alpha x_1[|n| - 1] + \beta x_2[|n| - 1] \\ &= \alpha y_1[n] + \beta y_2[n] \end{aligned}$$

Hence, the system is **linear**.

- **Time invariance:** Consider $y'[n]$ as output corresponding to $x'[n] = x[n - n_0]$, we have

$$\begin{aligned} y'[n] &= x'[|n| - 1] \\ &= x[|n| - 1 - n_0] \end{aligned}$$

However, $y[n - n_0] = x[|n - n_0| - 1] \neq y'[n]$, so the system is **time-varying**.

- **Causality**

Since the function requires possible future values (e.g. when $n=-1$, $y[-1] = x[0]$), the system is **non-causal**.

d) $y[n] = x[3n]$

- **Linearity:**

Consider $x_1[n]$, $x_2[n]$ as two inputs and their corresponding outputs as $y_1[n]$, $y_2[n]$. For another input $x_3[n] = \alpha x_1[n] + \beta x_2[n]$,

$$\begin{aligned} y_3[n] &= x_3[3n] \\ &= \alpha x_1[3n] + \beta x_2[3n] \\ &= \alpha y_1[n] + \beta y_2[n] \end{aligned}$$

Hence, the system is **linear**.

- **Time invariance:** Consider $y'[n]$ as output corresponding to $x'[n] = x[n - n_0]$, we have

$$\begin{aligned} y'[n] &= x'[3n] \\ &= x[3n - n_0] \end{aligned}$$

However, $y[n - n_0] = x[3(n - n_0)] \neq y'[n]$, so the system is **time-varying**.

- **Causality**

Since the function requires possible future values (e.g. when $n=1$, $y[1]$ requires value of $x[3]$), the system is **non-causal**.

e) $y[n] = \frac{x[n]}{x[2]}$

- **Linearity:**

This can be verified easily with homogeneity check only. Suppose $x_1[n] = \alpha x[n]$ is the input of $y_1[n]$

$$\begin{aligned} y_1[n] &= \frac{\alpha x[n]}{\alpha x[2]} \\ &= \frac{x[n]}{x[2]} \\ &\neq \alpha \frac{x[n]}{x[2]} = \alpha y[n] \end{aligned}$$

Hence, the system is **non-linear**.

- **Time invariance:**

Consider $y'[n]$ as output corresponding to $x'[n] = x[n - n_0]$, we have

$$\begin{aligned} y'[n] &= \frac{x'[n]}{x'[2]} \\ &= \frac{x[n - n_0]}{x[2 - n_0]} \end{aligned}$$

However, $y[n - n_0] = \frac{x[n - n_0]}{x[2]} \neq y'[n]$, so the system is **time-varying**

- **Causality**

Since the function requires possible future values (e.g. when $n=1$, $y[1]$ requires value of $x[2]$), the system is **non-causal**.

Grading: 30 points

- +2 each for correctly identifying and reasoning linearity (+1 if reasoning is partially correct)
- +2 each for correctly identifying and reasoning time-invariance/varying (+1 if reasoning is partially correct)
- +2 each for correctly identifying and reasoning causality (+1 if reasoning is partially correct)

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Solution:

Defining input-output pairs $x_1[n]$ and $y_1[n] = h[n] * x_1[n]$, $x_2[n]$ and $y_2[n] = h[n] * x_2[n]$, $x_3[n] = \alpha x_1[n] + \beta x_2[n]$ and $y_3[n] = h[n] * x_3[n]$, linearity can be shown if the output $y_3[n] = \alpha y_1[n] + \beta y_2[n]$.

Substituting this into the convolution definition,

$$\begin{aligned}
 y_3[n] &= h[n] * x_3[n] = \sum_{k=-\infty}^{\infty} h[n-k]x_3[k] \\
 &= \sum_{k=-\infty}^{\infty} h[n-k](\alpha x_1[k] + \beta x_2[k]) \\
 &= \alpha \sum_{k=-\infty}^{\infty} h[n-k]x_1[k] + \beta \sum_{k=-\infty}^{\infty} h[n-k]x_2[k] \\
 &= \alpha(h[n] * x_1[n]) + \beta(h[n] * x_2[n]) \\
 &= \alpha y_1[n] + \beta y_2[n],
 \end{aligned}$$

confirming that such a system must be linear.

Time invariance can be established by defining input-output pair $x'[n]$ and $y'[n]$ and output $y'[n]$ to delayed input $x'[n] = x[n - n_0]$ and showing that $y'[n] = y[n - n_0]$. Inserting this into the above definition,

$$\begin{aligned}
 y'[n] &= h[n] * x'[n] = \sum_{k=-\infty}^{\infty} h[n-k]x'[k] \\
 &= \sum_{k=-\infty}^{\infty} h[n-k]x[k - n_0] \\
 &= \sum_{k'=-\infty}^{\infty} h[n - (k' + n_0)]x[k'], \text{ where } k' = k - n_0 \\
 &= \sum_{k'=-\infty}^{\infty} h[(n - n_0) - k']x[k'],
 \end{aligned}$$

and

$$y[n - n_0] = \sum_{k=-\infty}^{\infty} h[(n - n_0) - k]x[k] = y'[n],$$

so such a system is time-invariant.

Grading: 10 points

- +5 for correctly identifying and reasoning linearity (+3 if reasoning is partially correct) (It's also accepted to use the distribution property for convolution mentioned in lecture.)
- +5 for correctly identifying and reasoning time-invariance/varying (+3 if reasoning is partially correct)

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Solution:

Using the hint, we can express a Delta as unit steps as below:

$$\delta[n] = u[n] - u[n - 1]$$

Using the definition:

$$g[n] = h[n] * u[n]$$

We can write hint as:

$$h[n] * \delta[n] = h[n] * (u[n] - u[n-1])$$

$$h[n] = g[n] - g[n-1]$$

For an LSI system:

$$y[n] = x[n] * h[n]$$

Substituting in the previous result:

$$y[n] = x[n] * (g[n] - g[n-1])$$

Grading: 10 points

- +3 Writes delta in terms of unit steps
- +7 Correct result

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Solution:

Notice that $\delta[n] = u[n] - u[n-1]$, and the signal contains a unit step. We can use linearity and shift-invariance to construct a pulse convoluted with the system's unit pulse response.

Since $y[n] = h[n] * x[n]$, we simply substitute the input and output and we get

$$\frac{1}{3^n} u[n-1] = h[n] * \frac{1}{2^n} u[n]. \quad (1)$$

Now we start to create a pulse on the right-hand side by first shifting

$$\frac{1}{3^{n-1}} u[n-2] = h[n] * \frac{1}{2^{n-1}} u[n-1],$$

and then multiply by $\frac{1}{2}$

$$\frac{1}{2 \cdot 3^{n-1}} u[n-2] = h[n] * \frac{1}{2 \cdot 2^{n-1}} u[n-1]. \quad (2)$$

Now subtract eq.(2) from eq.(1) so that we get $\delta[n]$ in the convolution,

$$\begin{aligned} \frac{1}{3^n} u[n-1] - \frac{1}{2 \cdot 3^{n-1}} u[n-2] &= h[n] * \frac{1}{2 \cdot 2^{n-1}} (u[n] - u[n-1]) \\ &= h[n] * \frac{1}{2^n} \delta[n] \\ &= h[n] * \delta[n] \\ &= h[n]. \end{aligned}$$

Therefore, the system's unit pulse response is $h[n] = \frac{1}{3^n} (u[n-1] - \frac{3}{2} u[n-2])$.

Grading: 15 points

- +15 for the correct answer
- -5 for using z-transform to solve the problem
- -1 each minor math error

Solution:

a) $x[n] = \{-1, 0, 1\}$, $h[n] = \{1, 2, 3, 4, 5\}$

Notice that $x[n]$ only has two nonzero elements, hence, write the convolution sum in the following form to get a simple expression with only two nonzero terms:

$$\begin{aligned}
 x[n] * h[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\
 &= x[-1]h[n+1] + x[1]h[n-1] \\
 &= h[n-1] - h[n+1] \\
 &= \{0, 1, 2, 3, 4, 5\} - \{1, 2, 3, 4, 5\} \\
 &= \{-1, -2, -2, -2, -2, 4, 5\}
 \end{aligned}$$

Alternative solution: Notice that $x[n] = \delta[n-1] - \delta[n+1]$, hence:

$$\begin{aligned}
 x[n] * h[n] &= (\delta[n-1] - \delta[n+1]) * h[n] \\
 &= h[n-1] - h[n+1] \\
 &= \dots
 \end{aligned}$$

b) $x[n] = 5^{-n}u[n]$, $h[n] = \{1, 0, 2\}$

Following the logic in part a, this time write the convolution sum in the alternative form:

$$\begin{aligned}
 x[n] * h[n] &= \sum_{k=-\infty}^{\infty} h[k]x[n-k] \\
 &= x[n] + 2x[n-2] \\
 &= 5^{-n}u[n] + 2 \cdot 5^{-n+2}u[n-2] \quad (*) \\
 &= 5^{-n}u[n] + 50 \cdot 5^{-n}u[n-2] \quad (*) \\
 &= \delta[n] + 0.2\delta[n-1] + 51 \cdot 5^{-n}u[n-2] \quad (*)
 \end{aligned}$$

all (*) are acceptable solutions.

c) $x[n] = u[n]$, $h[n] = n(u[n] - u[n-3])$

Simplify $h[n]$ as: $h[n] = \delta[n-1] + 2\delta[n-2]$. (HW1 - Problem 6b).

$$\begin{aligned}
 x[n] * h[n] &= u[n] * (\delta[n-1] + 2\delta[n-2]) \\
 &= u[n-1] + 2u[n-2]
 \end{aligned}$$

d) $x[n] = (-1)^{(-n)}u[n], h[n] = e^{(-n)}u[n]$

$$\begin{aligned}
x[n] * h[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\
&= \sum_{k=-\infty}^{\infty} (-1)^{-k}u[k]e^{-n+k}u[n-k] \\
&= \sum_{k=0}^{\infty} (-1)^{-k}e^{-n+k}u[n-k] \\
&= \sum_{k=0}^n (-1)^{-k}e^{-n+k} = \sum_{k=0}^n (-1)^k e^{-n+k} \\
&= \boxed{e^{-n} \sum_{k=0}^n (-e)^k} \quad (*)
\end{aligned}$$

Note that this sum is zero for $n < 0$ since the lower limit of the sum will be higher than the upper limit, which leads to an empty sum. Nevertheless, we can explicitly state this condition with a piecewise function definition, or alternatively utilizing the unit step function:

$$\begin{aligned}
x[n] * h[n] &= \begin{cases} e^{-n} \sum_{k=0}^n (-e)^k & n \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (*) \\
&= \boxed{u[n]e^{-n} \sum_{k=0}^n (-e)^k} \quad (*)
\end{aligned}$$

The expression can be reduced further using the geometric series finite sum formula:

$$\sum_{k=0}^n r^k = \frac{1 - r^{n+1}}{1 - r} \quad , \quad n \geq 0$$

Applying this to the summation we get:

$$\begin{aligned}
x[n] * h[n] &= u[n]e^{-n} \sum_{k=0}^n (-e)^k \\
&= u[n]e^{-n} \frac{1 - (-e)^{n+1}}{1 - (-e)} \\
&= \boxed{u[n] \frac{e^{-n} + e(-1)^n}{1 + e}} \quad (*)
\end{aligned}$$

all (*) are acceptable solutions.

e) $x[n] = 0.5^n u[n]$, $h[n] = 2^{(-n)} u[-n]$

$$\begin{aligned}
 x[n] * h[n] &= \sum_{k=-\infty}^{\infty} x[k] h[n-k] \\
 &= \sum_{k=-\infty}^{\infty} 0.5^k u[k] 0.5^{n-k} u[-n+k] \\
 &= \sum_{k=0}^{\infty} 0.5^n u[-n+k] \\
 &= \underbrace{0.5^n}_{>0} \underbrace{\sum_{k=0}^{\infty} u[-n+k]}_{=\infty \text{ (infinite sum of shifted unit steps)}} \\
 &= \boxed{\infty}
 \end{aligned}$$

Grading: 25 points

- +5 for each correct part (-1 each minor math error)

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Solution:

For an LSI system, the input-output relationship is written by the convolution sum:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k] \quad (3)$$

$$= \dots + h[-2]x[n+2] + h[-1]x[n+1] + h[0]x[n] + h[1]x[n-1] + h[2]x[n-2] + \dots \quad (4)$$

Let's assume that the system is causal, then $y[n]$ should not depend on any $x[n+k]$ for $k > 0$ (value of the output at a given time should not depend on the future input values). This implies that $h[k] = 0$ for $k < 0$ (see Eq. (3)).

Now, assume that $h[k] = 0$ for $k < 0$. Again, through Eq. (3), we see that the output $y[n]$ does not depend on any input $x[n+k]$ for $k > 0$, which means that the system is causal.

Grading: 10 points

- +5 for causality implies $h[k] = 0$ for $k < 0$
- +5 for $h[k] = 0$ for $k < 0$ implies causality